JUSTIN JOE MATHEW

B.Tech CSE Undergrad @ Amal Jyothi · Software Prototyping, Web Systems & Drone Tech Enthusiast

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TECHNICAL SKILLS

Working Knowledge	Python, JavaScript, TypeScript, C (Academic Core), HTML5, CSS3, Tailwind CSS
Familiar with / Exploring	C++, Kotlin, Dart, SQL, Node.js Basics
Web & Prototyping Tools	React, Next.js, Vite, Git & GitHub, Linux / Bash Shell, Arduino (Uno Q), Figma, Vercel
Data, Simulation & APIs	NumPy (Vectorized Simulation), FastF1 Telemetry API, REST APIs, JSON Data Streams
Languages (Spoken)	English (Fluent), Malayalam (Native), Hindi (Fluent)

FEATURED ENGINEERING PROJECTS

Formula 1 Monte Carlo Race Strategy Simulator

Live Demo ↗ · GitHub ↗

Python · NumPy · FastF1 API · React

- Architected a vectorized Python simulation tool using **NumPy arrays** to execute **10,000 stochastic race runs** in milliseconds, evaluating tyre degradation curves and safety-car probabilities.
- Integrated real-world Grand Prix telemetry feeds via the **FastF1 API** to calibrate stint wear rates and fuel-burn lap deltas.
- Built an interactive **React web dashboard** deploying real-time stint degradation graphs and optimal pit-window matrices.

AroggyaGram-3 — Community Crisis Triage Web Platform

1st Place Winner — ACM Vibefest 2026 · GitHub ↗

TypeScript · Next.js · WebSockets · Rapid Prototyping

- Engineered a responsive web application prototype designed to streamline emergency triage reporting for underserved communities.
- Prototyped incoming distress signal classification, parsing urgency levels and routing incident coordinates to verified local field responders.
- Secured **1st Place at ACM Vibefest Hackathon 2026** for rapid system design, live functional demo, and practical crisis impact.

Captain Lakshmi: Invicta — Interactive Web Monograph

Web Design Competition Entry · Live Demo ↗

Canvas 2D · React 19 · Vite · Hardware-Accelerated Scrubber

- Designed and developed an interactive biographical digital monograph chronicling Dr. Lakshmi Sahgal (INA) for a competitive web design event.
- Implemented a smooth **Canvas 2D frame-scrubber engine** cycling through 3,100+ frames at 60 FPS synchronized with responsive prose.
- Engineered with modern editorial typography, responsive CSS layout hierarchy, and minimal latency media buffering.

Autonomous Search & Rescue Drone Concept (SIH 2026 Proposal)

Smart India Hackathon 2026 Shortlist

Drone Systems Research · Hardware Architecture · Arduino Uno Q

- Researched and authored a disaster-response UAV systems proposal addressing aerial search-and-rescue under zero-cloud, communication blackout conditions.
- Incorporated hardware edge architecture insights gained from hands-on volunteering with the **Arduino Uno Q board** at the **Blankspace Hackathon (powered by Arduino)**.
- Built on 3+ years of active independent research, UAV literature exploration, and continuous technical mentorship from drone community resource persons.

HONORS, HACKATHONS & RECOGNITION

- **1st Place Winner** — **ACM Vibefest Hackathon 2026** (ACM-W Chapter, March 2026) — 24-hour rapid product prototyping & system architecture.
- **1st Place Winner** — **AI Club Ideathon 2026** (Amal Jyothi College of Engineering, Feb 2026) — Algorithmic community triage concept.
- **Stage 1 Winner** — **Idea From Industry Challenge Season 2** (MakerHub IEDC · BIRAC · MoE, Jan 2026) — Smart transit telemetry & fare concept.
- **National Shortlist** — **Smart India Hackathon (SIH 2026)** (Ministry of Education Innovation Cell & AICTE) — Search & Rescue UAV systems proposal.
- **Double 3rd Prize (Podium)** — **Azure 2026** (Feb 2026) — 3rd Prize in Extempore English & 3rd Prize in JamSpace (Just A Minute) impromptu speaking.
- **Participant** — **NASA Space Apps Challenge 2025** (Aerospace open data challenge) & **K-DISC YIP 8.0** (Govt. of Kerala).

EDUCATION & BACKGROUND

Amal Jyothi College of Engineering (Autonomous) — B.Tech in Computer Science & Engineering — Kottayam, Kerala | 2025 – 2029

Fatimagiri Higher Secondary School — CBSE Class XII (85.60%) — Nilambur, Kerala | 2025

> **Multicultural Adaptability & Versatility:** Educated across 5 schools spanning **Dubai (UAE), New Delhi, and multiple regions of Kerala**. Fluent in English, Malayalam, and Hindi. Naturally adaptable, communicative, and comfortable integrating quickly into diverse teams.